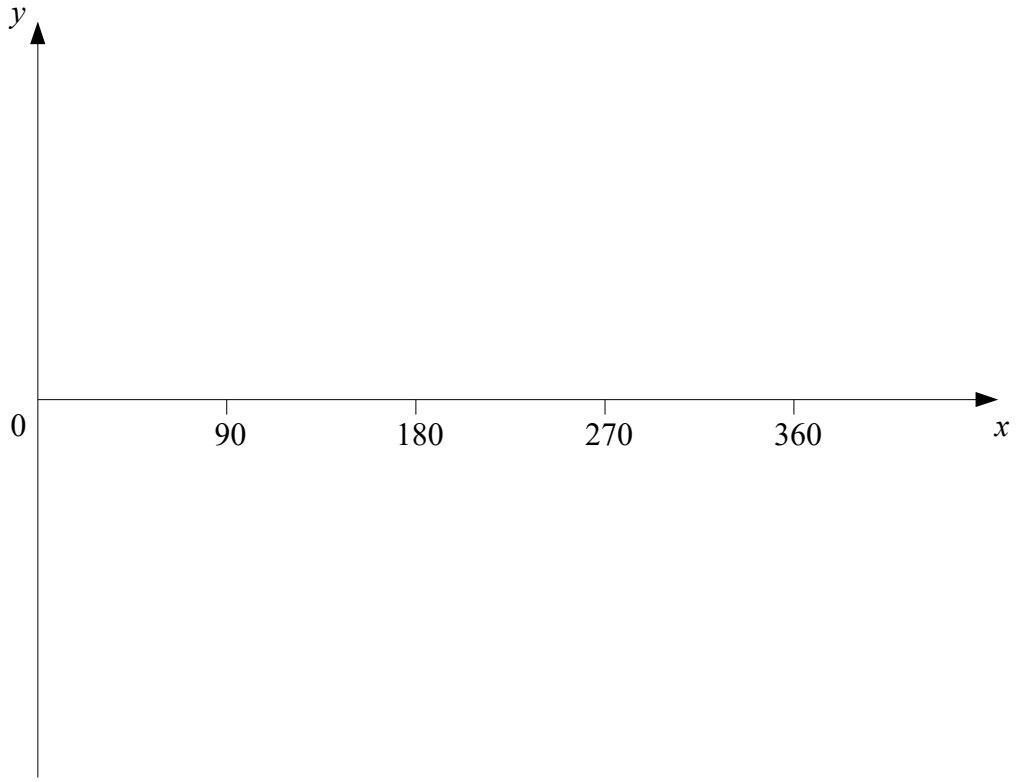
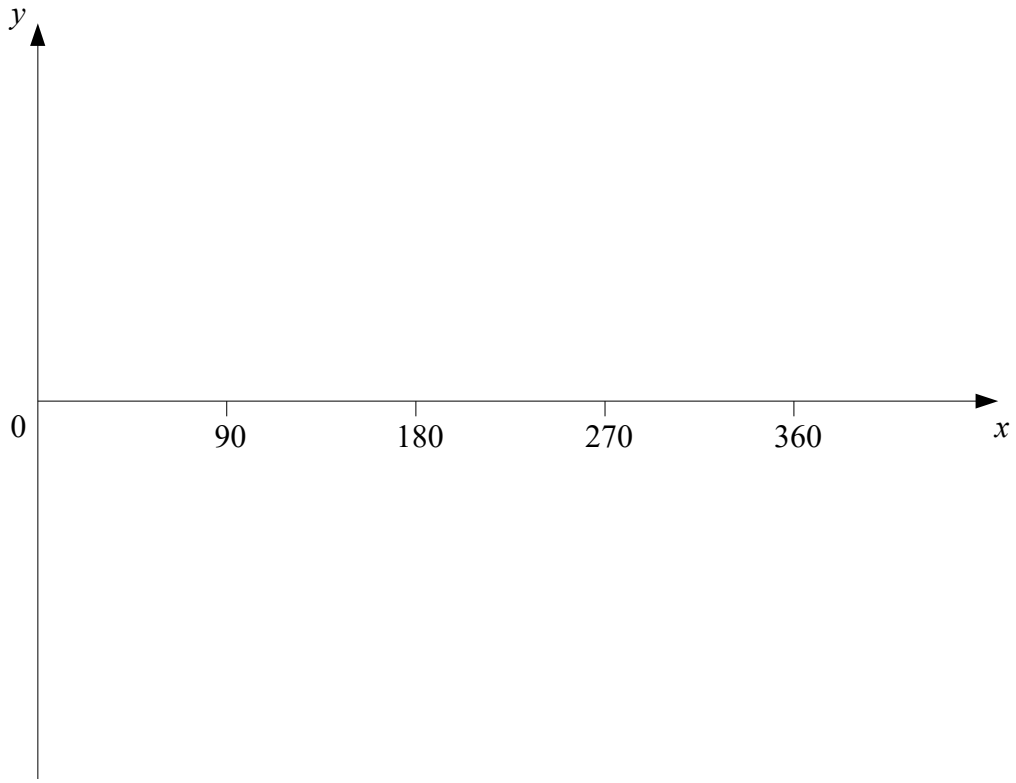
1 Sketch the graph of $y = \sin x°$ for $0 \leq x \leq 360$



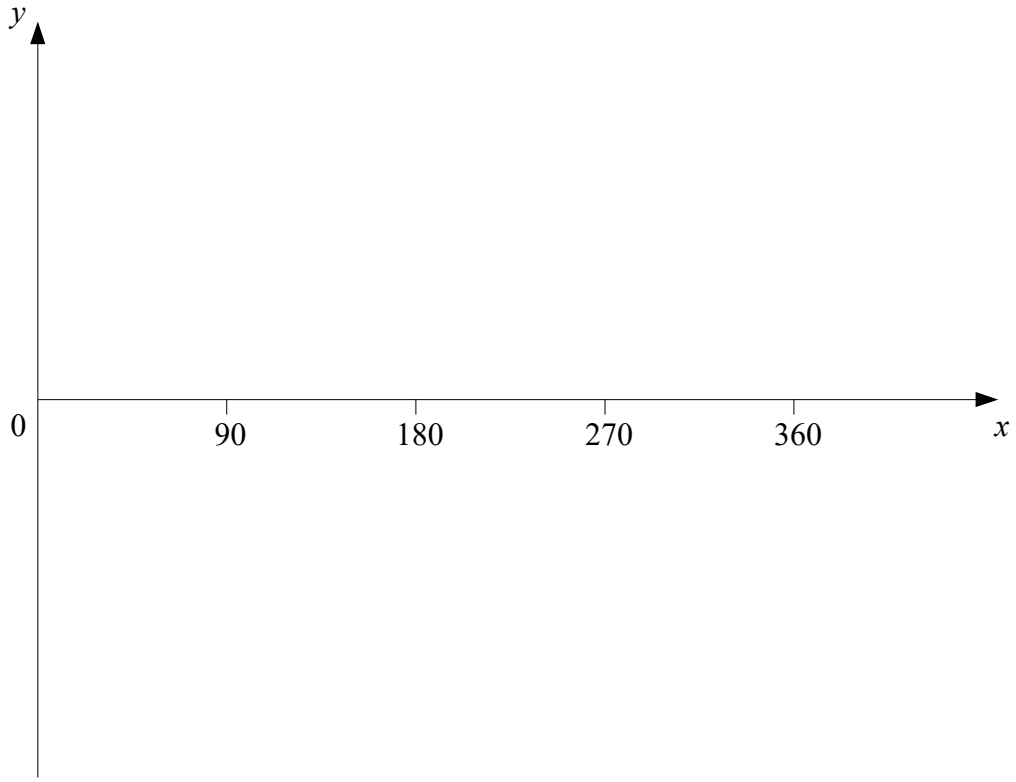
(Total for Question 1 is 2 marks)

2 Sketch the graph of $y = \tan x°$ for $0 \leq x \leq 360$



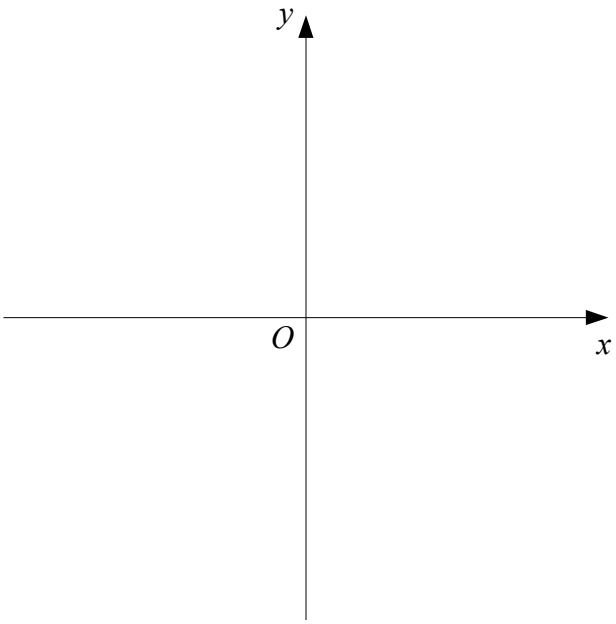
(Total for Question 2 is 2 marks)

3 Sketch the graph of $y = \cos x°$ for $0 \leq x \leq 360$



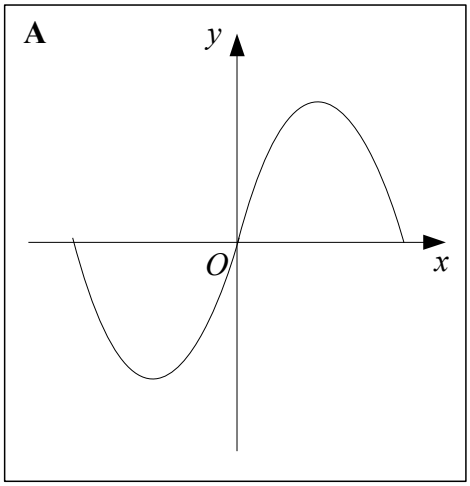
(Total for Question 3 is 2 marks)

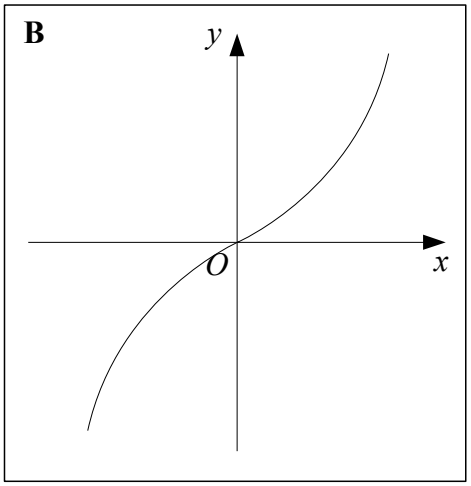
4 On the grid, sketch the curve with equation $y = 2^x$
Give the coordinates of any points of intersection with the axes.

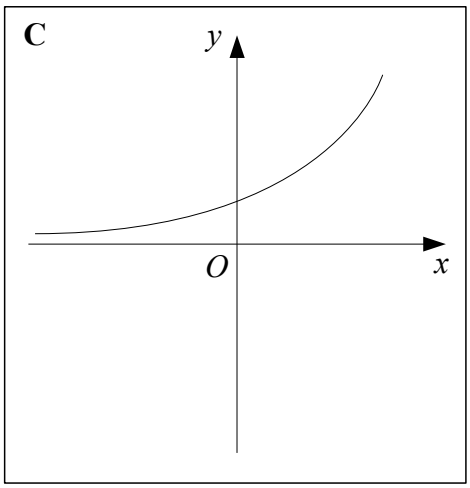


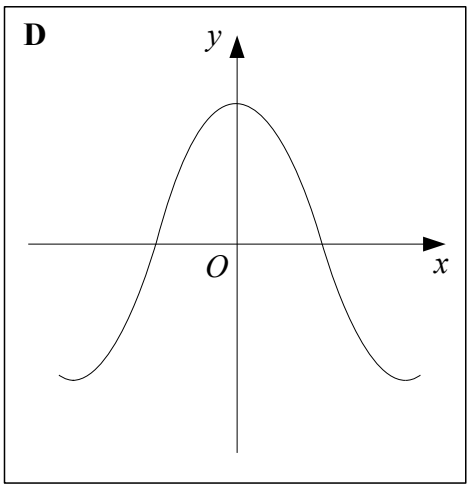
(Total for Question 4 is 2 marks)

5 Here are four graphs







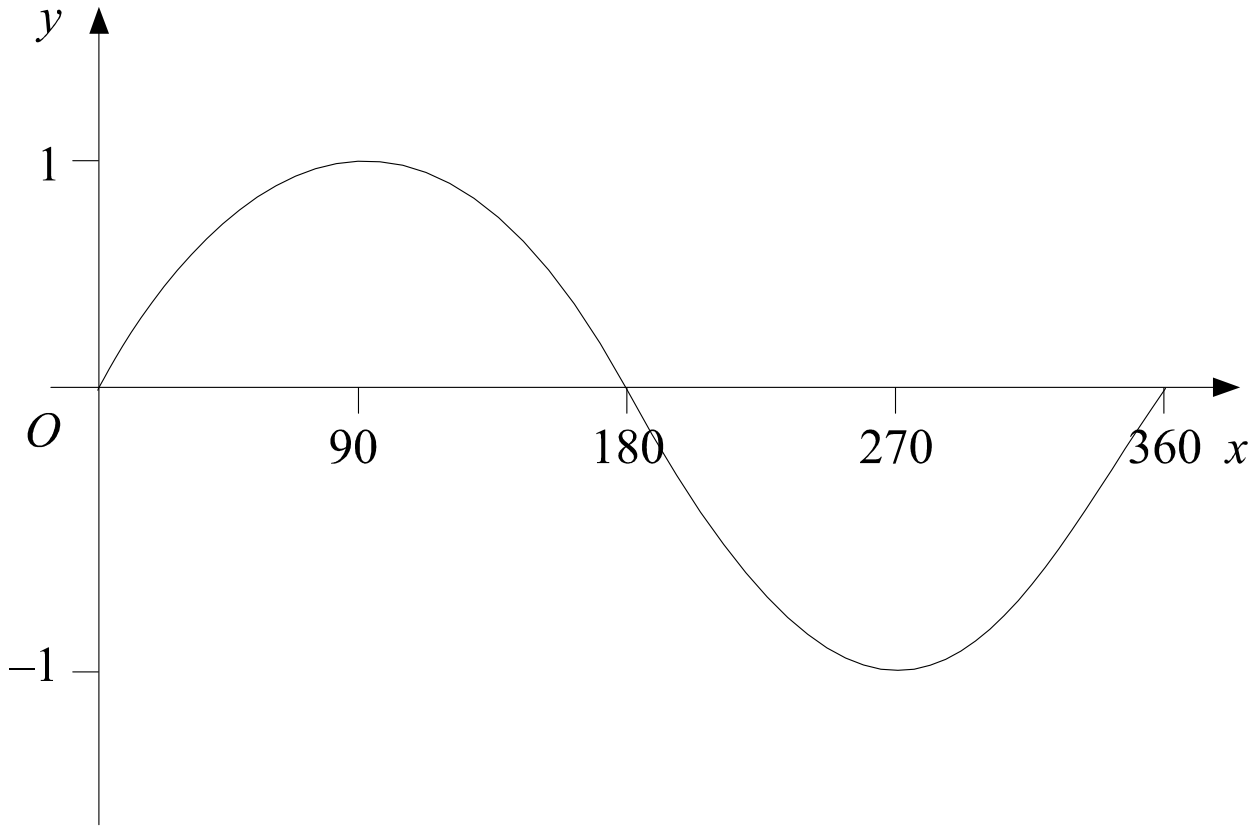


In the table below, match each equation with the letter of its graph.

Equation	Letter of Graph
$y = \sin x$	
$y = 2^x$	
$y = x^3$	
$y = \cos x$	

(Total for Question 5 is 2 marks)

6 Here is a sketch of the curve $y = \sin x°$ for $0 \leq x \leq 360$



Given that $\sin 30° = \frac{1}{2}$ write down the value of:

i) $\sin 150°$

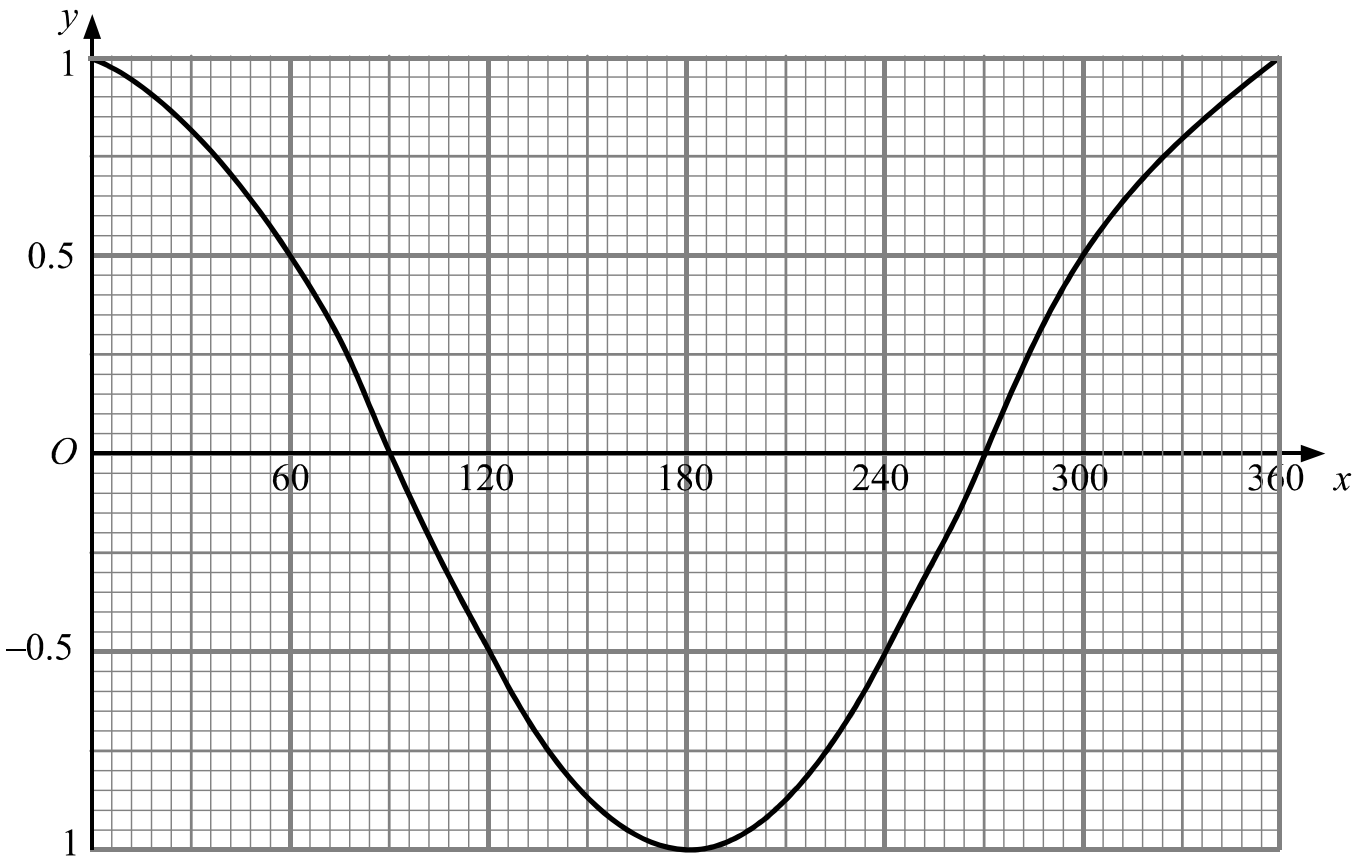
.....
(1)

ii) $\sin 330°$

.....
(1)

(Total for Question 6 is 2 marks)

7 Here is a sketch of the curve $y = \cos x°$ for $0 \leq x \leq 360$



Use the graph to find estimates of the solutions, in the interval $0 \leq x \leq 360$, of the equation:

i) $\cos(x) = -0.4$

.....

(2)

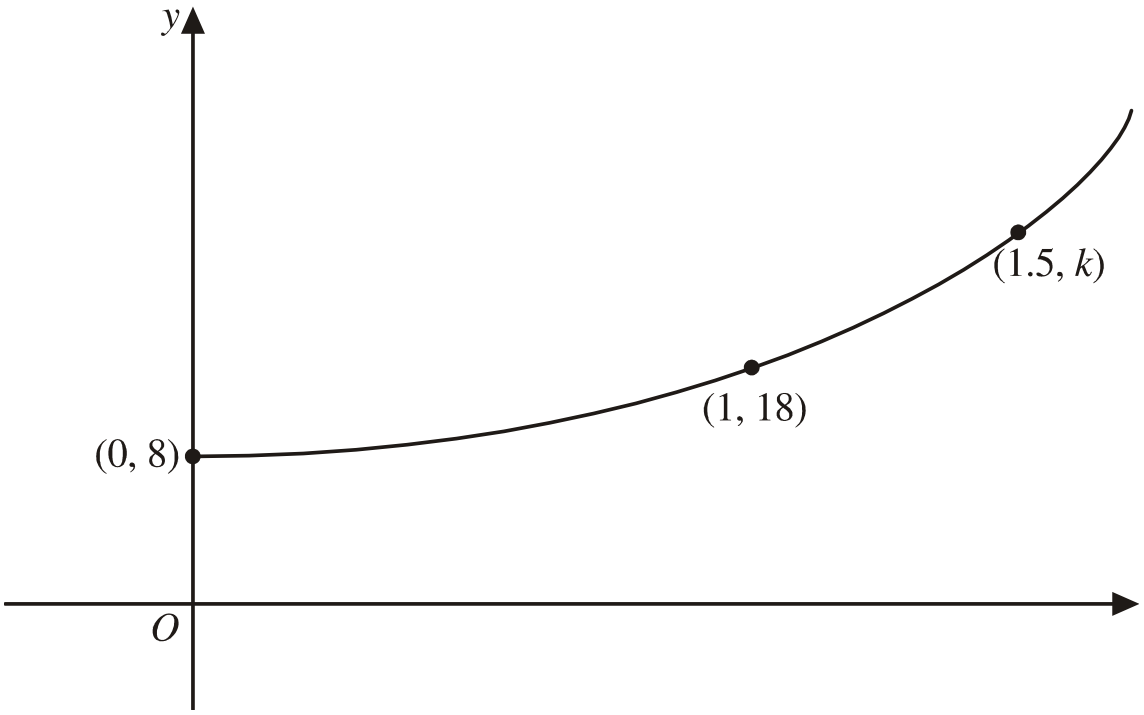
ii) $4\cos(x) = 3$

.....

(2)

(Total for Question 7 is 4 marks)

8 This sketch shows part of the graph with equation $y = pq^x$ where p and q are constants.



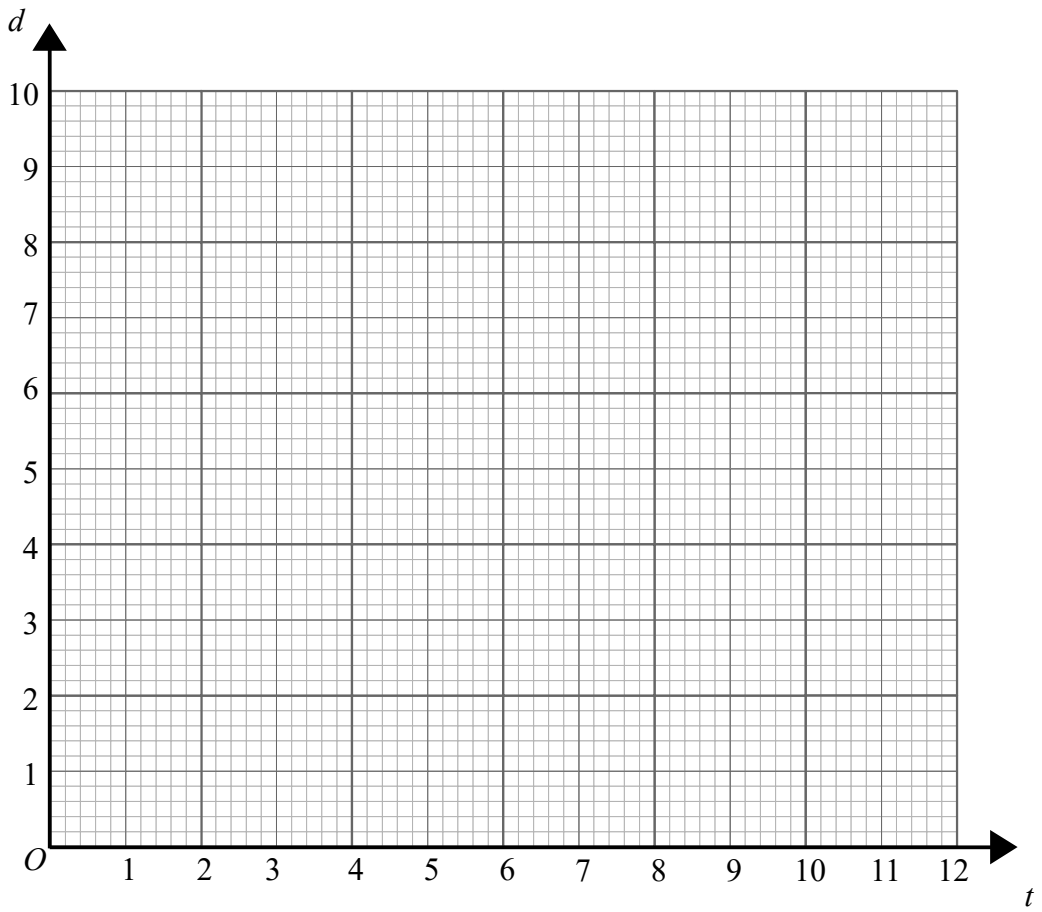
The points with coordinates (0, 8), (1, 18) and (1.5, k) lie on the graph.

Calculate the values of p, q and k.

(Total for Question 8 is 6 marks)

9 The depth of water, d metres, at the entrance to a harbour is given by the formula:
$d = 5 - 4\sin(30t)$, where t is the time in hours after midnight on one day.

(a) On the axes below, draw the graph of d against t for $0 \leq t \leq 12$ **(4)**



(b) Find the two values of t, where $0 \leq t \leq 24$, when the depth is least.

..... and .....

(2)

(Total for Question 9 is 6 marks)